Near-field singularity extraction for rooftop basis functions on uniform rectangular grids

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1 Introduction

This note continues the bachelor thesis [1]. There the Galerkin matrix of the electric field integral equation (EFIE) with rooftop basis functions on a uniform rectangular grid is assembled from three Toeplitz generators. Their entries are four-dimensional integrals of the Green's function

$$g(R) = \frac{\mathrm{e}^{-\mathrm{i}kR}}{R}, \qquad R = \|\boldsymbol{r} - \boldsymbol{r}'\|, \tag{1}$$

over pairs of cells, weighted by the linear profiles of the rooftops (the factor $1/(4\pi)$ is kept outside). They are integrated with a tensor Gauss–Legendre (GL) rule of order q, and the singularity is extracted on the coincident cell only (Entry 2.1 in Section 2.6 and the appendix of [1]).

Cells that share an edge or only a vertex meet $R = 0$ as well, on the boundary of the integration domain. There the GL error does not decrease with the order: the generator entries keep a relative error of about 10^{-3} for cells of $\lambda/10$ at every order between 3 and 16, and extraction on the coincident cell alone changes nothing measurable. The note therefore extracts the non-smooth terms of (1),

$$\frac{1}{R}, \qquad -\frac{k^2}{2}R, \qquad \frac{k^4}{24}R^3, \tag{2}$$

on all nine touching cells and gives the closed forms of every static integral this needs.

Three terms are enough. With the three terms the entries reach 10^{-13}–10^{-14} at the order 12–16, and the computed current is at the rounding floor of double precision. A fourth term (R^5) would act below that floor, so no further extraction is needed.

The way to the closed forms. Section 2 expresses the three generators through seven cell-pair moments, so that only integrals over pairs of cells with linear weights are left. Section 3 splits the Green's function into the three extracted terms and a smooth remainder; the remainder stays with the GL rule, and the extracted terms leave the static moments with the kernels R^{-1}, R and R^3. Section 4 reduces these four-dimensional integrals to two dimensions, and Section 5 gives the antiderivatives of the resulting integrands. Section 6 puts them together: every static moment of a touching cell pair is a short sum of antiderivative values. The note is written as a manual to the implementation; all closed forms are listed in Appendices A and B in the form of the thesis appendix.

Conventions. The time convention is $\mathrm{e}^{\mathrm{i}\omega t}$. The notation follows the thesis [1].

2 Setting: cells, rooftops and moments

Definition 2.1 (Uniform grid). Let $a, b > 0$ be the cell sizes in the x and y directions. For $(m, n) \in \mathbb{Z}^2$ the cell is

$$\Omega_{mn} = [ma, (m+1)a] \times [nb, (n+1)b] \subset \mathbb{R}^2. \tag{3}$$

A point $\boldsymbol{r} = (x, y) \in \Omega_{mn}$ has the local coordinates

$$\xi(\boldsymbol{r}) = \frac{x}{a} - m \in [0, 1], \qquad \eta(\boldsymbol{r}) = \frac{y}{b} - n \in [0, 1]. \tag{4}$$

For an offset $\boldsymbol{d} = (d_x, d_y) \in \mathbb{Z}^2$ we write $\Omega_{\boldsymbol{d}} = \Omega_{d_x d_y}$. The two cells $\Omega_{\mathbf{0}}$ and $\Omega_{\boldsymbol{d}}$ share at least one point if and only if $\boldsymbol{d} \in \mathcal{T} = \{-1, 0, 1\}^2$. We call these offsets touching (Figure 1): $\boldsymbol{d} = \mathbf{0}$ is the coincident cell, the four offsets $(\pm 1, 0)$, $(0, \pm 1)$ share an edge and the four offsets $(\pm 1, \pm 1)$ a vertex.

Definition 2.2 (Rooftop functions). The x-directed rooftop $\boldsymbol{\Psi}^x_{mn}$ is supported on $\Omega_{mn} \cup \Omega_{m+1,n}$ and

$$\boldsymbol{\Psi}^x_{mn}(\boldsymbol{r}) = \hat{\boldsymbol{x}} \cdot \begin{cases} \xi(\boldsymbol{r}), & \boldsymbol{r} \in \Omega_{mn} \quad \text{(rising wedge)}, \\ 1 - \xi(\boldsymbol{r}), & \boldsymbol{r} \in \Omega_{m+1,n} \quad \text{(falling wedge)}. \end{cases} \tag{5}$$

Its normal component on the common edge is one, and its divergence is $+1/a$ on Ω_{mn} and $-1/a$ on $\Omega_{m+1,n}$. The y-directed rooftop $\boldsymbol{\Psi}^y_{mn}$ is defined in the same way with η, b and $\Omega_{m,n+1}$.

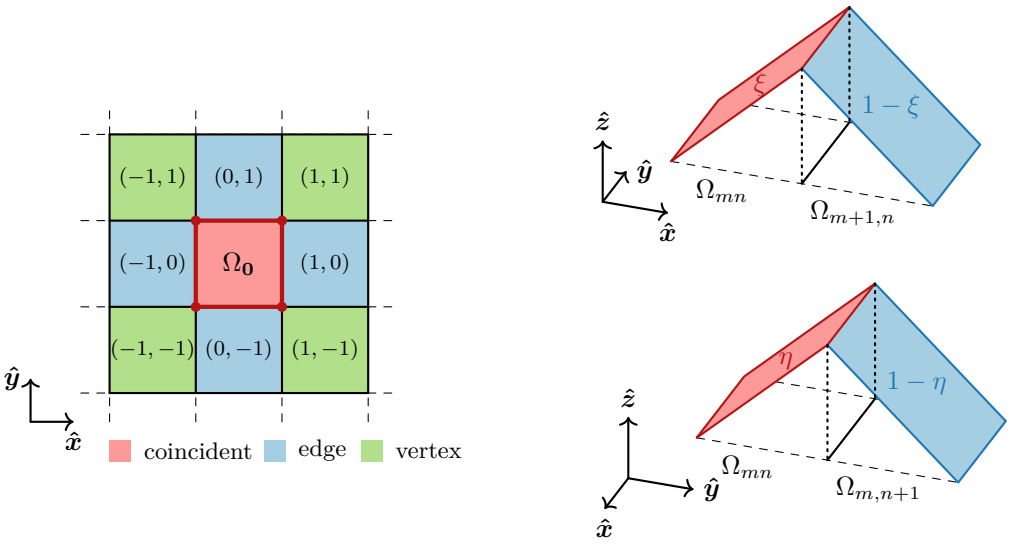


Figure 1: Left: the reference cell $\Omega_{\mathbf{0}}$ and its eight touching neighbours $\Omega_{\boldsymbol{d}}$, labelled by the offset $\boldsymbol{d} \in \mathcal{T}$. The points with $R = 0$ that the two cells have in common are drawn in red: all of $\Omega_{\mathbf{0}}$ for the coincident cell, an edge for an edge neighbour and a vertex for a vertex neighbour. Right: the x-directed rooftop $\boldsymbol{\Psi}^x_{mn}$ (top), split into its rising wedge ξ on Ω_{mn} and its falling wedge $1-\xi$ on $\Omega_{m+1,n}$, and the y-directed rooftop $\boldsymbol{\Psi}^y_{mn}$ (bottom) with the wedges η on Ω_{mn} and $1 - \eta$ on $\Omega_{m,n+1}$.

Entry 2.1 (Galerkin entries). With (1), the Galerkin entry of the EFIE between rooftops $\boldsymbol{\Psi}_p$ and $\boldsymbol{\Psi}_q$ is [1]

$$\begin{aligned} Z_{pq} = \frac{\mathrm{i} Z_0}{4\pi k} \Big(& k^2 \iint \boldsymbol{\Psi}_p(\boldsymbol{r}) \cdot \boldsymbol{\Psi}_q(\boldsymbol{r}') \, g(R) \, \mathrm{d}A' \, \mathrm{d}A \\ & - \iint \nabla \cdot \boldsymbol{\Psi}_p(\boldsymbol{r}) \, \nabla' \cdot \boldsymbol{\Psi}_q(\boldsymbol{r}') \, g(R) \, \mathrm{d}A' \, \mathrm{d}A \Big). \end{aligned} \tag{6}$$

By translation invariance the entries depend only on the offset of the two rooftops. They are generated by

$$A^{xx}(\boldsymbol{d}) = \iint \boldsymbol{\Psi}^x_{\mathbf{0}} \cdot \boldsymbol{\Psi}^x_{\boldsymbol{d}} \, g, \qquad A^{yy}(\boldsymbol{d}) = \iint \boldsymbol{\Psi}^y_{\mathbf{0}} \cdot \boldsymbol{\Psi}^y_{\boldsymbol{d}} \, g, \qquad \Phi(\boldsymbol{d}) = \int_{\Omega_{\mathbf{0}}} \int_{\Omega_{\boldsymbol{d}}} g. \tag{7}$$

The divergences are piecewise constant, so the scalar-potential parts of all blocks are finite differences of Φ; for example $\iint \nabla\cdot\boldsymbol{\Psi}^x_{\mathbf{0}}\,\nabla'\cdot\boldsymbol{\Psi}^x_{\boldsymbol{d}}\,g = a^{-2}\big(2\Phi(\boldsymbol{d})-\Phi(\boldsymbol{d}+\boldsymbol{e}_x)-\Phi(\boldsymbol{d}-\boldsymbol{e}_x)\big)$.

Definition 2.3 (Cell-pair moments). Let $K:(0,\infty)\to\mathbb{C}$ satisfy $K(\|\cdot\|)\in L^1_{\mathrm{loc}}(\mathbb{R}^2)$. For $\boldsymbol{d}\in\mathbb{Z}^2$ and $\alpha,\beta\in\{0,1\}$ define

$$\begin{aligned} M^{[K]}_{\alpha\beta}(\boldsymbol{d}) &= \frac{1}{(ab)^2}\int_{\Omega_{\mathbf{0}}}\int_{\Omega_{\boldsymbol{d}}} \xi(\boldsymbol{r})^\alpha\,\xi(\boldsymbol{r}')^\beta\,K\big(\|\boldsymbol{r}-\boldsymbol{r}'\|\big)\,\mathrm{d}A(\boldsymbol{r}')\,\mathrm{d}A(\boldsymbol{r}),\\ N^{[K]}_{\alpha\beta}(\boldsymbol{d}) &= \frac{1}{(ab)^2}\int_{\Omega_{\mathbf{0}}}\int_{\Omega_{\boldsymbol{d}}} \eta(\boldsymbol{r})^\alpha\,\eta(\boldsymbol{r}')^\beta\,K\big(\|\boldsymbol{r}-\boldsymbol{r}'\|\big)\,\mathrm{d}A(\boldsymbol{r}')\,\mathrm{d}A(\boldsymbol{r}), \end{aligned} \tag{8}$$

where $\xi(\boldsymbol{r}')$ and $\eta(\boldsymbol{r}')$ are the local coordinates in $\Omega_{\boldsymbol{d}}$. For $K(R)=R^p$ we write $M^{(p)}_{\alpha\beta}(\boldsymbol{d})$ and $N^{(p)}_{\alpha\beta}(\boldsymbol{d})$. Since $N_{00}=M_{00}$, there are seven moments, M_{00}, M_{10}, M_{01}, M_{11}, N_{10}, N_{01} and N_{11}.

The moments carry all the geometry. The next lemma lists their symmetries.

Lemma 2.1 (Symmetries of the moments). Let K be admissible, $\boldsymbol{d}=(d_x,d_y)\in\mathbb{Z}^2$ and $\alpha,\beta\in\{0,1\}$. Then:

(i) $M^{[K]}_{\alpha\beta}(d_x,-d_y)=M^{[K]}_{\alpha\beta}(d_x,d_y)$;

(ii) reflection in x:

$$\begin{aligned} M^{[K]}_{00}(-d_x,d_y) &= M^{[K]}_{00}(\boldsymbol{d}),\\ M^{[K]}_{10}(-d_x,d_y) &= \big(M^{[K]}_{00}-M^{[K]}_{10}\big)(\boldsymbol{d}),\\ M^{[K]}_{01}(-d_x,d_y) &= \big(M^{[K]}_{00}-M^{[K]}_{01}\big)(\boldsymbol{d}),\\ M^{[K]}_{11}(-d_x,d_y) &= \big(M^{[K]}_{00}-M^{[K]}_{10}-M^{[K]}_{01}+M^{[K]}_{11}\big)(\boldsymbol{d}); \end{aligned} \tag{9}$$

(iii) $N^{[K]}_{\alpha\beta}(d_x,d_y;a,b)=M^{[K]}_{\alpha\beta}(d_y,d_x;b,a)$ (the cell sizes written explicitly);

(iv) $M^{[K]}_{\alpha\beta}(\boldsymbol{d})=M^{[K]}_{\beta\alpha}(-\boldsymbol{d})$.

Proof. The kernel depends only on $\|\boldsymbol{r}-\boldsymbol{r}'\|$, which is invariant under reflections, translations and the interchange of $\boldsymbol{r}$ and $\boldsymbol{r}'$.

(i) The reflection $y\mapsto b-y$ maps $\Omega_{\mathbf{0}}$ onto itself and $\Omega_{(d_x,-d_y)}$ onto $\Omega_{(d_x,d_y)}$. It leaves both local x-coordinates unchanged.

(ii) The reflection $x\mapsto a-x$ maps $\Omega_{\mathbf{0}}$ onto itself with $\xi\mapsto 1-\xi$, and $\Omega_{(-d_x,d_y)}$ onto $\Omega_{(d_x,d_y)}$ with $\xi'\mapsto 1-\xi'$. Indeed, $[-d_xa,(1-d_x)a]$ goes to $[d_xa,(d_x+1)a]$ and $x'/a+d_x\mapsto 1-(x'/a+d_x)$. Expand $(1-\xi)^\alpha(1-\xi')^\beta$.

(iii) Exchange the roles of x and y.

(iv) Interchange $\boldsymbol{r}\leftrightarrow\boldsymbol{r}'$ and translate by $-\boldsymbol{d}$; the cell $\Omega_{\boldsymbol{d}}$ becomes $\Omega_{\mathbf{0}}$ and $\Omega_{\mathbf{0}}$ becomes $\Omega_{-\boldsymbol{d}}$.

□

Proposition 2.1 (Generators from moments). For $\boldsymbol{d}=(d_x,d_y)$ with $d_x\ge 0$,

$$\begin{aligned} \frac{A^{xx}(\boldsymbol{d})}{(ab)^2} &= M_{11}(\boldsymbol{d})+\big(M_{00}-M_{10}-M_{01}+M_{11}\big)(\boldsymbol{d})\\ &\quad+\big(M_{10}-M_{11}\big)(\boldsymbol{d}+\boldsymbol{e}_x)+\big(M_{01}-M_{11}\big)(\boldsymbol{d}-\boldsymbol{e}_x), \end{aligned} \tag{10}$$

with $M = M^{[g]}$. For $d_x = 0$ the last term is to be read as

$$\big(M_{01} - M_{11}\big)(-1, d_y) = \big(M_{10} - M_{11}\big)(1, d_y). \tag{11}$$

Furthermore $\Phi(\boldsymbol{d}) = (ab)^2 M_{00}^{[g]}(\boldsymbol{d})$, and A^{yy} is given by (10) with N and $\boldsymbol{e}_y$ in place of M and $\boldsymbol{e}_x$.

Proof. Split both rooftops into their wedges, as in Definition (2.2). The reference rooftop $\boldsymbol{\Psi}_{\mathbf{0}}^x$ has the rising wedge ξ on $\Omega_{\mathbf{0}}$ and the falling wedge $1-\xi$ on $\Omega_{\boldsymbol{e}_x}$. The rooftop $\boldsymbol{\Psi}_{\boldsymbol{d}}^x$ has the rising wedge ξ' on $\Omega_{\boldsymbol{d}}$ and the falling wedge $1-\xi'$ on $\Omega_{\boldsymbol{d}+\boldsymbol{e}_x}$. Both are x-directed, so the dot product is the product of the profiles. The four wedge pairs of Figure 2 give

pair	cells	offset	weight	moment
1, rising–rising	$\Omega_{\mathbf{0}}$, $\Omega_{\boldsymbol{d}}$	$\boldsymbol{d}$	$\xi\xi'$	M_{11}
2, falling–falling	$\Omega_{\boldsymbol{e}_x}$, $\Omega_{\boldsymbol{d}+\boldsymbol{e}_x}$	$\boldsymbol{d}$	$(1-\xi)(1-\xi')$	$M_{00} - M_{10} - M_{01} + M_{11}$
3, rising–falling	$\Omega_{\mathbf{0}}$, $\Omega_{\boldsymbol{d}+\boldsymbol{e}_x}$	$\boldsymbol{d}+\boldsymbol{e}_x$	$\xi(1-\xi')$	$M_{10} - M_{11}$
4, falling–rising	$\Omega_{\boldsymbol{e}_x}$, $\Omega_{\boldsymbol{d}}$	$\boldsymbol{d}-\boldsymbol{e}_x$	$(1-\xi)\xi'$	$M_{01} - M_{11}$

with the moments taken at the offset of the pair (for pair 4 after the translation by $-\boldsymbol{e}_x$). The normalization $(ab)^{-2}$ in Definition (2.3) accounts for the factor $(ab)^2$. For $d_x = 0$ the offset $\boldsymbol{d} - \boldsymbol{e}_x = (-1, d_y)$ is negative. Lemma (2.1)(ii) gives

$$\begin{aligned}(M_{01} - M_{11})(-1, d_y) &= (M_{00} - M_{01})(1, d_y) - (M_{00} - M_{10} - M_{01} + M_{11})(1, d_y) \\ &= (M_{10} - M_{11})(1, d_y).\end{aligned} \tag{12}$$

The statement on Φ is the definition with $\alpha = \beta = 0$. The y-direction is identical. □

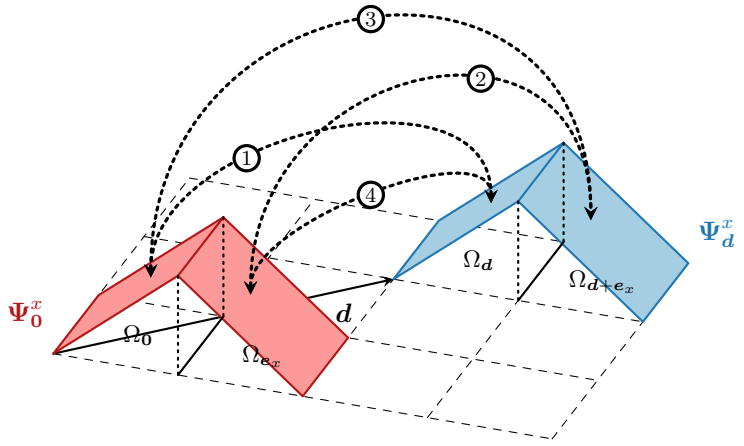


Figure 2: The four wedge pairs of the rooftops $\boldsymbol{\Psi}_{\mathbf{0}}^x$ and $\boldsymbol{\Psi}_{\boldsymbol{d}}^x$ in Proposition (2.1), drawn for $\boldsymbol{d} = (2, 2)$, numbered as in its proof. Every pair is a pair of cells at one offset, weighted by a product of two linear profiles.

The generators of all three Toeplitz blocks need the seven moments on the offsets $d_x \geq -1$, $d_y \geq 0$, and by Lemma (2.1) on $d_x, d_y \geq 0$ only.

3 Singularity extraction on the touching cells

Proposition 3.1 (Non-smooth terms of the Green's function). For $R \geq 0$,

$$g(R) = \frac{\mathrm{e}^{-\mathrm{i}kR}}{R} = \sum_{n=0}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} R^{n-1} = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!} R^{2\ell-1} \; - \; \mathrm{i} \sum_{\ell=0}^{\infty} \frac{(-1)^\ell k^{2\ell+1}}{(2\ell+1)!} R^{2\ell}. \quad (13)$$

The series converge absolutely and locally uniformly for $R > 0$; the second one is an entire function of R^2. As functions of $\boldsymbol{z} = \boldsymbol{r} - \boldsymbol{r}' \in \mathbb{R}^2$ the even powers $R^{2\ell} = \|\boldsymbol{z}\|^{2\ell}$ are polynomials, while the odd powers $R^{2\ell-1}$ are not smooth at $\boldsymbol{z} = \boldsymbol{0}$.

Proof. Insert the exponential series and separate the even and odd n; this is Proposition 2.2 in Section 2.6 of [1].

□

Definition 3.1 (Extraction of order N). For $N \in \mathbb{N}$ let

$$S_N(R) = \sum_{\ell=0}^{N-1} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!} R^{2\ell-1} = \frac{1}{R} - \frac{k^2}{2} R + \frac{k^4}{24} R^3 - \dots, \qquad g_N = g - S_N. \quad (14)$$

Table 1 lists the orders: $N = 1$ is the classical extraction of the static singularity [2], $N = 2$ the two-term extraction of [1] (Entry 2.1 in Section 2.6), there written as

$$g = g_{\mathrm{s}} + \frac{1}{4\pi} R^{-1} - \frac{k^2}{8\pi} R, \quad (15)$$

and $N = 3$ the one of this note. For every $N \geq 1$, g_N extends continuously to $R = 0$ with $g_N(0) = -\mathrm{i}k$.

Lemma 3.1 (Regularity of the remainder). For $N \geq 1$ the function $\boldsymbol{z} \mapsto g_N(\|\boldsymbol{z}\|)$ is real-analytic on $\mathbb{R}^2 \setminus \{\boldsymbol{0}\}$ and belongs to $C^{2N-2,1}$ on every bounded set: its derivatives up to the order $2N-2$ are continuous, and those of the order $2N-2$ are Lipschitz. It is not of class C^{2N-1} unless $k = 0$.

Proof. By (13),

$$g_N(R) = E(R^2) + R^{2N-1} F(R^2), \quad (16)$$

with the entire functions

$$E(s) = -\mathrm{i} \sum_{\ell \geq 0} \frac{(-1)^\ell k^{2\ell+1}}{(2\ell+1)!} s^\ell, \qquad F(s) = \sum_{\ell \geq N} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!} s^{\ell-N}. \quad (17)$$

The function $E(\|\boldsymbol{z}\|^2)$ is real-analytic on $\mathbb{R}^2$.

The function $\|\boldsymbol{z}\|^{2N-1} = \left(\|\boldsymbol{z}\|^2\right)^{N-1} \|\boldsymbol{z}\|$ is smooth away from the origin and positively homogeneous of degree $2N-1$. Hence its derivatives of order $m \leq 2N-2$ are homogeneous of degree $2N-1-m \geq 1$. They are therefore continuous at the origin (with the value zero), and those of order $2N-2$, being homogeneous of degree one, are Lipschitz. The derivatives of order $2N-1$ are homogeneous of degree zero and not constant, so they are bounded but discontinuous at the origin.

Multiplication by the analytic factor $F(\|\boldsymbol{z}\|^2)$ preserves the class $C^{2N-2,1}$ on bounded sets. Since $F(0) = (-1)^N k^{2N}/(2N)! \neq 0$ for $k \neq 0$, the term $F(0)\,\|\boldsymbol{z}\|^{2N-1}$ is present and the class is sharp.

□

On a touching cell pair the integrand of a moment meets $\boldsymbol{z} = \mathbf{0}$, in the interior of the four-dimensional domain for the coincident cell and on its boundary for a shared edge or vertex. A tensor GL rule integrates polynomials exactly, so its error on such a pair is governed by the regularity of Lemma (3.1) (Table 1).

Table 1: The extraction orders and the regularity of the remainder on a touching cell pair.

N	extracted S_N	used in	remainder g_N
0	—		unbounded
1	$1/R$	[2]	Lipschitz
2	$1/R - k^2R/2$	[1]	$C^{2,1}$
3	$1/R - k^2R/2 + k^4R^3/24$	this note	$C^{4,1}$

The error then decreases algebraically in the order q, and the faster, the larger N. The lemma is not a quadrature error estimate; for that theory see [3]. On the cell pairs that do not touch, g is real-analytic on a neighbourhood of the domain, and the GL error decreases exponentially.

Corollary 3.1 (Exact decomposition of the touching moments). For every $N \geq 1$, $\boldsymbol{d} \in \mathcal{T}$ and $\alpha, \beta \in \{0, 1\}$,

$$M^{[g]}_{\alpha\beta}(\boldsymbol{d}) = M^{[g_N]}_{\alpha\beta}(\boldsymbol{d}) + \sum_{\ell=0}^{N-1} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!}\, M^{(2\ell-1)}_{\alpha\beta}(\boldsymbol{d}), \tag{18}$$

and likewise for the N-moments. The first term is evaluated with GL; the static moments $M^{(p)}_{\alpha\beta}(\boldsymbol{d})$, $p \in \{-1, 1, 3\}$, are given in closed form by Theorem (6.1).

Proof. The moments are linear in K, and every term is absolutely integrable: the kernels $R^{2\ell-1}$ with $\ell \geq 0$ have $2\ell - 1 \geq -1 > -2$, and g_N is bounded. □

What has to be computed. By Lemma (2.1), the static moments of the four offsets

$$\boldsymbol{d} \in \{(0,0),\ (1,0),\ (0,1),\ (1,1)\} \tag{19}$$

determine all nine touching offsets. These are the coincident cell, the two kinds of shared edge and the shared vertex. For $N = 3$ we need, for each of them, the four x-moments $M_{00}, M_{10}, M_{01}, M_{11}$ and the three y-moments, with $p \in \{-1, 1, 3\}$. The y-moments follow from the x-moments by Lemma (2.1)(iii).

4 Reduction to two dimensions

The integrand of a moment depends on the four coordinates only through the two differences $x - x'$ and $y - y'$, and the weights are polynomials. Integrating out the "position along the diagonal" in each direction leaves a two-dimensional integral with piecewise polynomial weights. The coincident-cell transformation of [1] (Proposition 2.3 in Section 2.6, regions I–IV), which follows [4], is the special case $\boldsymbol{d} = \mathbf{0}$.

Lemma 4.1 (One-dimensional correlation). Let $\varphi, \psi \in L^\infty(0,1)$, $\delta \in \mathbb{R}$, and let $h : \mathbb{R} \to \mathbb{C}$ be locally integrable. Then

$$\int_0^1\!\!\int_0^1 \varphi(\xi)\,\psi(\tau)\,h(\xi-\tau-\delta)\,\mathrm{d}\tau\,\mathrm{d}\xi = \int_{-1}^{1} W_{\varphi\psi}(\nu)\,h(\nu-\delta)\,\mathrm{d}\nu,$$
$$W_{\varphi\psi}(\nu) = \int_{\max(0,\nu)}^{\min(1,1+\nu)} \varphi(\xi)\,\psi(\xi-\nu)\,\mathrm{d}\xi, \tag{20}$$

and both integrals converge absolutely.

Proof. The linear map $(\xi,\tau) \mapsto (\xi,\nu)$, $\nu = \xi - \tau$, has determinant one. It maps the unit square onto the parallelogram (Figure 3)

$$\{(\xi,\nu) :\ -1 \le \nu \le 1,\ \max(0,\nu) \le \xi \le \min(1,1+\nu)\}, \tag{21}$$

because $0 \le \tau = \xi - \nu \le 1$ is equivalent to $\nu \le \xi \le 1+\nu$.

By Tonelli's theorem applied to $|\varphi|\,|\psi|\,|h|$, the left-hand side is absolutely convergent, since

$$\int_0^1\!\!\int_0^1 |\varphi(\xi)\,\psi(\tau)\,h(\xi-\tau-\delta)|\,\mathrm{d}\tau\,\mathrm{d}\xi \le \|\varphi\|_\infty\,\|\psi\|_\infty \int_{-1}^{1} (1-|\nu|)\,|h(\nu-\delta)|\,\mathrm{d}\nu < \infty. \tag{22}$$

Fubini's theorem then allows the integration in ξ first, for fixed ν, which gives $W_{\varphi\psi}(\nu)$. □

Corollary 4.1 (Weights of the rooftop moments). For $\varphi(\xi) = \xi^\alpha$ and $\psi(\tau) = \tau^\beta$, $\alpha, \beta \in \{0,1\}$, write $W_{\alpha\beta} = W_{\varphi\psi}$. On $I_- = [-1,0]$ and $I_+ = [0,1]$ these are the polynomials

	$\nu \in I_-$	$\nu \in I_+$
W_{00}	$1+\nu$	$1-\nu$
W_{10}	$\tfrac{1}{2}(1+\nu)^2$	$\tfrac{1}{2}(1-\nu^2)$
W_{01}	$\tfrac{1}{2}(1-\nu^2)$	$\tfrac{1}{2}(1-\nu)^2$
W_{11}	$\tfrac{1}{6}(1+\nu)^2(2-\nu)$	$\tfrac{1}{6}(2-3\nu+\nu^3)$

(23)

They are continuous on $[-1,1]$, W_{00} and W_{11} are even, and $W_{01}(\nu) = W_{10}(-\nu)$.

Proof. For $\nu \in I_+$ the limits in (20) are $\nu \le \xi \le 1$:

$$W_{00} = \int_\nu^1 \mathrm{d}\xi = 1-\nu, \qquad W_{10} = \int_\nu^1 \xi\,\mathrm{d}\xi = \frac{1}{2}(1-\nu^2),$$
$$W_{01} = \int_\nu^1 (\xi-\nu)\,\mathrm{d}\xi = \frac{1}{2}(1-\nu)^2, \quad W_{11} = \int_\nu^1 \xi(\xi-\nu)\,\mathrm{d}\xi = \frac{1}{6}\big(2-3\nu+\nu^3\big). \tag{24}$$

For $\nu \in I_-$ they are $0 \le \xi \le 1+\nu$:

$$\begin{aligned} W_{00} &= \int_0^{1+\nu} \mathrm{d}\xi = 1+\nu, & W_{10} &= \int_0^{1+\nu} \xi \,\mathrm{d}\xi = \frac{1}{2}(1+\nu)^2, \\ W_{01} &= \int_0^{1+\nu} (\xi-\nu)\,\mathrm{d}\xi = \frac{1}{2}(1-\nu^2), & W_{11} &= \int_0^{1+\nu} \xi(\xi-\nu)\,\mathrm{d}\xi = \frac{1}{6}(1+\nu)^2(2-\nu). \end{aligned} \tag{25}$$

At $\nu = 0$ both sides give the values 1, 1/2, 1/2 and 1/3, so the weights are continuous. For the symmetries, the substitution $s = \xi + \nu$ in (20) gives $W_{\varphi\psi}(-\nu) = W_{\psi\varphi}(\nu)$. Hence $W_{01}(\nu) = W_{10}(-\nu)$, and W_{00}, W_{11} are even; for W_{11} compare

$$\frac{1}{6}(1-\nu)^2(2+\nu) = \frac{1}{6}\big(2 - 3\nu + \nu^3\big). \tag{26}$$

□

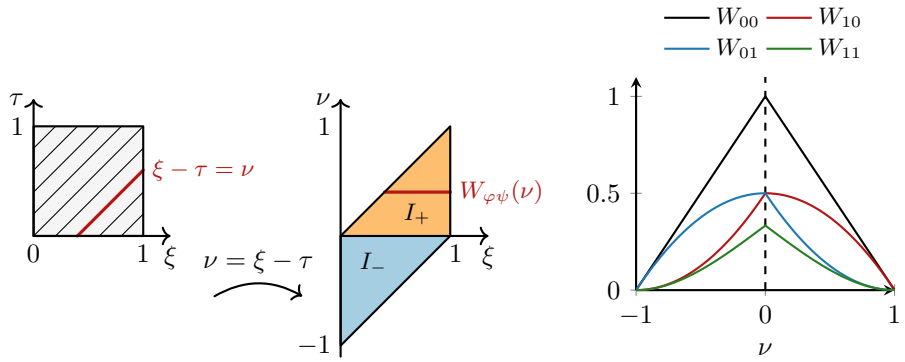


Figure 3: Left: the substitution $\nu = \xi - \tau$ of Lemma (4.1) maps the unit square onto a parallelogram; the line $\xi - \tau = \nu$ (red) becomes the horizontal segment over which $W_{\varphi\psi}(\nu)$ integrates. The two triangles $\nu \in I_-$ and $\nu \in I_+$ have different limits in ξ, which is why the weights change their polynomial form at $\nu = 0$. Right: the weights (23).

Theorem 4.1 (Two-dimensional representation). Let $p > -2$, $\boldsymbol{d} = (d_x, d_y) \in \mathbb{Z}^2$ and $\alpha, \beta \in \{0, 1\}$, and put

$$\rho_{\boldsymbol{d}}(\nu, \eta) = \sqrt{a^2(\nu - d_x)^2 + b^2(\eta - d_y)^2}. \tag{27}$$

Then

$$\begin{aligned} M^{(p)}_{\alpha\beta}(\boldsymbol{d}) &= \int_{-1}^{1}\int_{-1}^{1} W_{\alpha\beta}(\nu)\, W_{00}(\eta)\, \rho_{\boldsymbol{d}}(\nu,\eta)^p \,\mathrm{d}\eta\,\mathrm{d}\nu, \\ N^{(p)}_{\alpha\beta}(\boldsymbol{d}) &= \int_{-1}^{1}\int_{-1}^{1} W_{00}(\nu)\, W_{\alpha\beta}(\eta)\, \rho_{\boldsymbol{d}}(\nu,\eta)^p \,\mathrm{d}\eta\,\mathrm{d}\nu. \end{aligned} \tag{28}$$

The same holds for a general admissible kernel K, with $K(\rho_{\boldsymbol{d}})$ in place of $\rho^p_{\boldsymbol{d}}$.

Proof. Parametrize $\boldsymbol{r} = (a\xi_1, b\eta_1) \in \Omega_{\boldsymbol{0}}$ and $\boldsymbol{r}' = \big(a(d_x + \xi_2),\, b(d_y + \eta_2)\big) \in \Omega_{\boldsymbol{d}}$ with $\xi_1, \eta_1, \xi_2, \eta_2 \in [0, 1]$. Then $\mathrm{d}A'\,\mathrm{d}A = (ab)^2\,\mathrm{d}\eta_2\,\mathrm{d}\xi_2\,\mathrm{d}\eta_1\,\mathrm{d}\xi_1$, which cancels the normaliza-

tion of Definition (2.3), $\xi(\boldsymbol{r}) = \xi_1$, $\xi(\boldsymbol{r}') = \xi_2$, and

$$\|\boldsymbol{r} - \boldsymbol{r}'\| = \sqrt{a^2(\xi_1 - \xi_2 - d_x)^2 + b^2(\eta_1 - \eta_2 - d_y)^2}. \tag{29}$$

The moment is therefore the integral over the unit cube $[0,1]^4$

$$M^{(p)}_{\alpha\beta}(\boldsymbol{d}) = \int_0^1\int_0^1\int_0^1\int_0^1 \xi_1^\alpha\, \xi_2^\beta \left(a^2(\xi_1-\xi_2-d_x)^2 + b^2(\eta_1-\eta_2-d_y)^2\right)^{p/2} \mathrm{d}\eta_2\, \mathrm{d}\xi_2\, \mathrm{d}\eta_1\, \mathrm{d}\xi_1. \tag{30}$$

The integrand depends on the coordinates of the two directions only through the differences $\xi_1 - \xi_2$ and $\eta_1 - \eta_2$, and the weight depends on the x-coordinates only. The proof integrates out one difference after the other with Lemma (4.1), first in x and then in y.

First, the integral converges absolutely. The function $\boldsymbol{z} \mapsto \|\boldsymbol{z}\|^p$ is locally integrable on $\mathbb{R}^2$ for $p > -2$ (polar coordinates). Lemma (4.1), applied in each direction to the nonnegative kernel by Tonelli's theorem, gives

$$\int\!\!\int\!\!\int\!\!\int |\cdot| = \int_{-1}^1\int_{-1}^1 (1-|\nu|)(1-|\eta|)\, \rho_{\boldsymbol{d}}^p \le \int_{-1}^1\int_{-1}^1 \rho_{\boldsymbol{d}}^p < \infty. \tag{31}$$

Fubini's theorem therefore allows the four integrations of (30) in any order.

Second, the x-direction. Fix η_1 and η_2, and write $c = b(\eta_1 - \eta_2 - d_y)$ for the separation of the two points in y. The integrals over ξ_1 and ξ_2 in (30) are then

$$\int_0^1\int_0^1 \xi_1^\alpha\, \xi_2^\beta \left(a^2(\xi_1-\xi_2-d_x)^2 + c^2\right)^{p/2} \mathrm{d}\xi_2\, \mathrm{d}\xi_1. \tag{32}$$

This is the left-hand side of (20) with $\varphi(\xi) = \xi^\alpha$, $\psi(\tau) = \tau^\beta$, $\delta = d_x$ and the function of one real variable

$$h(s) = \left(a^2 s^2 + c^2\right)^{p/2}, \qquad s \in \mathbb{R}. \tag{33}$$

In the lemma h is evaluated at $s = \xi_1 - \xi_2 - d_x$. Then $a^2 s^2 = (x - x')^2$ is the squared separation of the two points in x, and $a^2 s^2 + c^2 = \|\boldsymbol{r} - \boldsymbol{r}'\|^2$. For $c \neq 0$ the function h is continuous, hence locally integrable. The pairs with $c = 0$, that is $\eta_1 - \eta_2 = d_y$, form a line in the (η_1, η_2)-square and do not contribute to the outer integral. The lemma, with the weight $W_{\varphi\psi} = W_{\alpha\beta}$ of Corollary (4.1), gives

$$\int_0^1\int_0^1 \xi_1^\alpha\, \xi_2^\beta \left(a^2(\xi_1-\xi_2-d_x)^2 + c^2\right)^{p/2} \mathrm{d}\xi_2\, \mathrm{d}\xi_1 = \int_{-1}^1 W_{\alpha\beta}(\nu) \left(a^2(\nu-d_x)^2 + c^2\right)^{p/2} \mathrm{d}\nu. \tag{34}$$

The two x-coordinates are replaced by their difference ν, and the weights ξ_1^α, ξ_2^β by the single weight $W_{\alpha\beta}(\nu)$. With $c = b(\eta_1 - \eta_2 - d_y)$ inserted back,

$$M^{(p)}_{\alpha\beta}(\boldsymbol{d}) = \int_{-1}^1 W_{\alpha\beta}(\nu) \left(\int_0^1\int_0^1 \left(a^2(\nu-d_x)^2 + b^2(\eta_1-\eta_2-d_y)^2\right)^{p/2} \mathrm{d}\eta_2\, \mathrm{d}\eta_1 \right) \mathrm{d}\nu, \tag{35}$$

where the ν-integration has been moved outside by Fubini's theorem.

Third, the y-direction. Fix $\nu \neq d_x$; the single value $\nu = d_x$ again does not contribute. The inner integral in the last display is the left-hand side of (20) once more, now in the variables (η_1, η_2) with $\varphi = \psi = 1$, $\delta = d_y$ and

$$h(s) = \left(a^2(\nu - d_x)^2 + b^2 s^2\right)^{p/2}, \qquad s \in \mathbb{R}, \tag{36}$$

where s now stands for $\eta_1 - \eta_2 - d_y$, so that $b^2 s^2 = (y - y')^2$ is the squared separation of the two points in y. For the weights 1 and 1 the weight of the lemma is W_{00}, so

$$\int_0^1 \int_0^1 \Big(a^2(\nu - d_x)^2 + b^2(\eta_1 - \eta_2 - d_y)^2 \Big)^{p/2} \, \mathrm{d}\eta_2 \, \mathrm{d}\eta_1 = \int_{-1}^{1} W_{00}(\eta) \, \rho_{\boldsymbol{d}}(\nu, \eta)^p \, \mathrm{d}\eta. \tag{37}$$

Inserting this into the previous display gives the first line of (28).

Finally, the N-moments carry the weights $\eta_1^\alpha \eta_2^\beta$ in the y-coordinates instead. The same three steps then give $W_{00}(\nu)$ in x and $W_{\alpha\beta}(\eta)$ in y, which is the second line of (28). Nothing in the argument uses the form of the power ρ^p beyond its local integrability, so a general admissible kernel K gives the same result with $K(\rho_{\boldsymbol{d}})$ in place of $\rho_{\boldsymbol{d}}^p$.

□

Remark. For $\boldsymbol{d} \in \mathcal{T}$ the singular point of $\rho_{\boldsymbol{d}}^p$, $p < 0$, is $(\nu, \eta) = (d_x, d_y)$. It is the centre, a midpoint of an edge or a corner of $[-1, 1]^2$ (Figure 5). In every case it lies on the lines $\nu \in \{-1, 0, 1\}$ and $\eta \in \{-1, 0, 1\}$, where the weights change their polynomial form. This is the structural fact behind the closed forms of Section 6.

5 Mixed antiderivatives

The antiderivatives below are functions of two lengths, which are written as a and b like in the thesis appendix. This does not clash with the cell sizes: the touching moments evaluate them only at the points (a, b), $(2a, b)$, $(a, 2b)$ and $(2a, 2b)$ (Section 6).

Definition 5.1 (Mixed antiderivatives). Let $Q = [0, \infty)^2$ be the closed first quadrant and Q° its interior. For $p > -2$, $m, n \in \mathbb{N}_0$ and $(a, b) \in Q$ put

$$G_{mn}^{(p)}(a, b) = \int_0^a \int_0^b u^m v^n \big(u^2 + v^2\big)^{p/2} \, \mathrm{d}v \, \mathrm{d}u. \tag{38}$$

For a function F on Q and a rectangle $A \times B \subset Q$ with $A = [a_0, a_1]$ and $B = [b_0, b_1]$, the corner difference of F over $A \times B$ is

$$\Delta_{A \times B} F = F(a_1, b_1) - F(a_0, b_1) - F(a_1, b_0) + F(a_0, b_0). \tag{39}$$

Proposition 5.1 (Properties of G). Let $p > -2$ and $m, n \in \mathbb{N}_0$.

(i) $G_{mn}^{(p)}$ is continuous on Q and vanishes on both axes: $G_{mn}^{(p)}(a, 0) = G_{mn}^{(p)}(0, b) = 0$.

(ii) For every rectangle $[a_0, a_1] \times [b_0, b_1] \subset Q$,

$$\int_{a_0}^{a_1} \int_{b_0}^{b_1} u^m v^n \big(u^2 + v^2\big)^{p/2} \, \mathrm{d}v \, \mathrm{d}u = \Delta_{[a_0, a_1] \times [b_0, b_1]} G_{mn}^{(p)}. \tag{40}$$

(iii) $\partial_a \partial_b G_{mn}^{(p)} = a^m b^n \big(a^2 + b^2\big)^{p/2}$ on Q°, and $G_{mn}^{(p)}$ is the only function in $C(Q) \cap C^2(Q^\circ)$ with this property that vanishes on both axes.

(iv) $G_{mn}^{(p)}(\lambda a, \lambda b) = \lambda^{m+n+p+2} \, G_{mn}^{(p)}(a, b)$ for $\lambda > 0$.

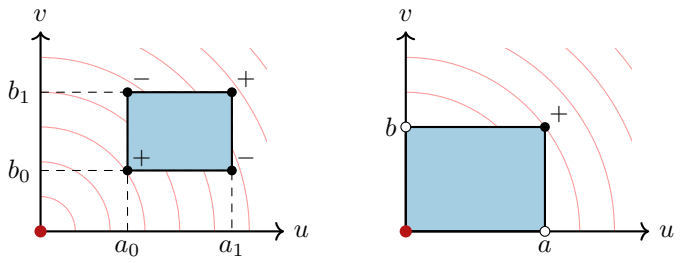


Figure 4: Left: the corner difference (40). The integral over the shaded rectangle is the alternating sum of $G^{(p)}_{mn}$ at its corners, each value being the integral over $[0,a]\times[0,b]$ (dashed). The circles are level lines of $\sqrt{u^2+v^2}$ around the singular point (red). Right: a piece that touches the singular point. Three of its corners lie on the axes, where $G^{(p)}_{mn}$ vanishes, and only $G^{(p)}_{mn}(a,b)$ remains.

Proof. (i) The integrand is locally integrable on Q because $p>-2$; continuity follows by dominated convergence, and the integral over a rectangle with a side of length zero is zero.

(ii) This is inclusion–exclusion (Figure 4): the rectangle $[a_0,a_1]\times[b_0,b_1]$ is $[0,a_1]\times[0,b_1]$ minus $[0,a_0]\times[0,b_1]$ and $[0,a_1]\times[0,b_0]$, plus their overlap $[0,a_0]\times[0,b_0]$.

(iii) The derivative follows from the fundamental theorem of calculus in each variable. If D is the difference of two such functions, $\partial_a\partial_b D=0$ gives $D(a,b)=e(a)+c(b)$ on Q°; letting $b\to0^+$ and $a\to0^+$ with $D=0$ on the axes shows that e and c are constants with $e+c=D(0,0)=0$.

(iv) Substitute $u=\lambda u'$ and $v=\lambda v'$ in (38).

□

Lemma 5.1 (One-dimensional integrals). Let $a\ge0$, $b>0$, $m\in\mathbb{N}_0$ and $q\in\{-1,1,3,5\}$, and let $R=\sqrt{a^2+b^2}$. The integrals

$$K^{(q)}_m(a)=\int_0^a u^m\bigl(u^2+b^2\bigr)^{q/2}\,\mathrm{d}u \tag{41}$$

are

$$K^{(q)}_0=\begin{cases}\log\left(\dfrac{R+a}{b}\right), & q=-1,\\[2mm] \dfrac{1}{2}\Bigl(aR+b^2\log\left(\dfrac{R+a}{b}\right)\Bigr), & q=1,\\[2mm] \dfrac{1}{8}\Bigl(2aR^3+3b^2aR+3b^4\log\left(\dfrac{R+a}{b}\right)\Bigr), & q=3,\\[2mm] \dfrac{1}{48}\Bigl(8aR^5+10b^2aR^3+15b^4aR+15b^6\log\left(\dfrac{R+a}{b}\right)\Bigr), & q=5,\end{cases} \tag{42}$$

$$K^{(q)}_1=\frac{R^{q+2}-b^{q+2}}{q+2}, \tag{43}$$

$$K^{(q)}_m=\frac{a^{m-1}R^{q+2}-(m-1)b^2K^{(q)}_{m-2}}{m+q+1},\qquad m\ge2, \tag{44}$$

and, for the logarithmic terms produced by the inner integrations,

$$\int_0^a u^m \log\left(\frac{\sqrt{u^2+b^2}+b}{u}\right)\mathrm{d}u = \frac{1}{m+1}\Big(a^{m+1}\log\left(\frac{R+b}{a}\right) + b\,K_m^{(-1)}(a)\Big). \tag{45}$$

The same formulas with a and b exchanged give the inner integrals $\int_0^b v^n(u^2+v^2)^{p/2}\,\mathrm{d}v$.

Proof. The base cases follow from

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}u}\log\left(\frac{\sqrt{u^2+b^2}+u}{b}\right) &= \big(u^2+b^2\big)^{-1/2},\\
\frac{\mathrm{d}}{\mathrm{d}u}\big(u^2+b^2\big)^{(q+2)/2} &= (q+2)\,u\,\big(u^2+b^2\big)^{q/2},\\
\frac{\mathrm{d}}{\mathrm{d}u}\Big(u\big(u^2+b^2\big)^{q/2}\Big) &= (q+1)\big(u^2+b^2\big)^{q/2} - qb^2\big(u^2+b^2\big)^{(q-2)/2};
\end{aligned} \tag{46}$$

the first one gives $K_0^{(-1)}$ and the second one (43). The last one gives the recursion

$$K_0^{(q)} = \frac{aR^q + qb^2K_0^{(q-2)}}{q+1}, \tag{47}$$

which produces (42) from $K_0^{(-1)}$.

For the recursion with $m \geq 2$, integrate from 0 to a

$$\frac{\mathrm{d}}{\mathrm{d}u}\big(u^{m-1}(u^2+b^2)^{(q+2)/2}\big) = (m-1)b^2u^{m-2}(u^2+b^2)^{q/2} + (m+q+1)u^m(u^2+b^2)^{q/2}. \tag{48}$$

The logarithmic terms follow by integration by parts with

$$\frac{\mathrm{d}}{\mathrm{d}u}\log\left(\frac{\sqrt{u^2+b^2}+b}{u}\right) = -\frac{b}{u\sqrt{u^2+b^2}}; \tag{49}$$

the boundary term $u^{m+1}\log\big((\sqrt{u^2+b^2}+b)/u\big)$ vanishes as $u \to 0^+$.

□

Proposition 5.2 (Closed forms). For $p \in \{-1,1,3\}$, $m \in \{0,1,2,3\}$, $n \in \{0,1\}$ and $(a,b) \in Q^\circ$,

$$\begin{aligned}
G_{mn}^{(p)}(a,b) = \frac{1}{D}\Bigg(&P_0 + P_1\sqrt{a^2+b^2} + P_2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ P_3\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg),
\end{aligned} \tag{50}$$

with an integer D and homogeneous integer polynomials $P_0,\dots,P_3$ in a and b, listed in Appendix A. These are all the antiderivatives the moments need; any other one follows from the symmetry $G_{nm}^{(p)}(b,a) = G_{mn}^{(p)}(a,b)$.

Proof. The formulas are obtained by integrating first in v and then in u with Lemma (5.1); every step produces only terms of the form (50).

□

Example. For $p = -1$, (45) gives directly

$$\begin{aligned} G_{00}^{(-1)}(a,b) &= \int_0^a \log\left(\frac{\sqrt{u^2+b^2}+b}{u}\right) \mathrm{d}u \\ &= a\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + b\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right), \end{aligned} \tag{51}$$

the classical potential integral of a uniformly charged rectangle at its corner.

6 Closed forms for the touching cells

Theorem (4.1) turns a moment into the two-dimensional integral (28) over $[-1,1]^2$, with the weights $W_{\alpha\beta}(\nu)$, $W_{00}(\eta)$ and the distance $\rho_{\boldsymbol{d}}(\nu,\eta)$. For a touching offset $\boldsymbol{d} = (d_x, d_y) \in \{0,1\}^2$ the distance vanishes at the singular point $(\nu,\eta) = (d_x, d_y)$, and this is the only point where the integrand is not smooth. The plan of this section is to cut $[-1,1]^2$ into pieces on which the weights are polynomials and to move the singular point of every piece to the origin. Each piece then becomes a rectangle in the quadrant Q of Section 5, with polynomial weights in front of $(u^2+v^2)^{p/2}$, and Proposition (5.1)(ii) integrates it exactly with the antiderivatives $G_{mn}^{(p)}$.

The cut is the one of Corollary (4.1): in each direction the interval $[-1,1]$ splits into $I_- = [-1,0]$ and $I_+ = [0,1]$, and on each of them the weights are polynomials. This gives the four pieces $I_s \times I_t$, $s,t \in \{-,+\}$, of Figure 5. The next lemma moves the singular point to the origin, one direction at a time.

Lemma 6.1 (The polynomial pieces). Let $d \in \{0,1\}$ be the offset in the x-direction, so that the singular value of ν is $\nu = d$. On each interval I_s, $s \in \{-,+\}$, introduce the distance from the singular value, measured in units of length,

$$u = a\,|\nu - d|\,. \tag{52}$$

On I_s the difference $\nu - d$ does not change sign, so $|\nu - d| = \sigma(\nu - d)$ with a fixed sign $\sigma \in \{-1,+1\}$, and the substitution is linear with the inverse $\nu = d + \sigma u/a$. Explicitly,

$$\begin{cases} d = 0: & u = -a\nu \in [0,a] \text{ on } I_-, \qquad u = a\nu \in [0,a] \text{ on } I_+, \\ d = 1: & u = a(1-\nu) \in [a,2a] \text{ on } I_-, \qquad u = a(1-\nu) \in [0,a] \text{ on } I_+. \end{cases} \tag{53}$$

In all four cases the singular value $\nu = d$ corresponds to $u = 0$. Every weight $W_{\alpha\beta}$ of (23), restricted to I_s and written in u, is a polynomial in u of degree at most three; these polynomials are listed in Table 2. The y-direction is the same with η, b, d_y and $v = b\,|\eta - d_y|$.

Proof. For $d = 0$ the difference $\nu - d = \nu$ lies in $[-1,0]$ on I_- and in $[0,1]$ on I_+, so $\sigma = -1$ and $\sigma = +1$, respectively, and u runs over $[0,a]$ on both. For $d = 1$ the difference $\nu - 1$ lies in $[-2,-1]$ on I_- and in $[-1,0]$ on I_+, so $\sigma = -1$ on both, and $u = a(1-\nu)$ runs over $[a,2a]$ and $[0,a]$. This gives (53). On I_s every weight is one of the polynomials of (23), of degree at most three in ν, and the linear substitution $\nu = d + \sigma u/a$ keeps it a polynomial of the same degree in u. Expanding gives Table 2.

□

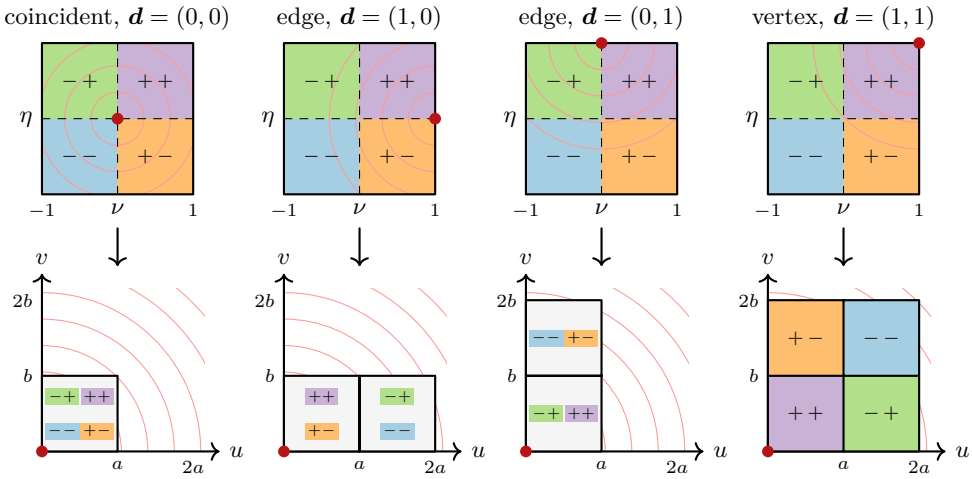


Figure 5: The integration regions of the four touching configurations ($a = b$). Top: the domain $[-1,1]^2$ of Theorem (4.1), split into the pieces $I_s \times I_t$, labelled $s\,t$; the singular point $(\nu,\eta) = (d_x, d_y)$ (red) is the centre, the midpoint of an edge or a corner, and the circles are level lines of $\rho_{\boldsymbol{d}}$. Bottom: the same pieces in the variables $u = \sigma_s a(\nu - d_x)$, $v = \sigma_t b(\eta - d_y)$ of Lemma (6.1). In a direction with $d = 0$ both pieces fold onto $[0,a]$; with $d = 1$ they lie side by side on $[0,a]$ and $[a,2a]$. The singular point is the origin, a corner of every piece that contains it.

Theorem 6.1 (Touching moments). Let $p \in \{-1,1,3\}$, $\boldsymbol{d} = (d_x, d_y) \in \{0,1\}^2$ and $\alpha,\beta \in \{0,1\}$. On the pieces $s,t \in \{-,+\}$ write, with Table 2,

$$W_{\alpha\beta}\big|_{I_s} = \sum_{m=0}^{3} c_m^s\, u^m, \qquad W_{00}\big|_{I_t} = \sum_{n=0}^{1} e_n^t\, v^n, \tag{54}$$

and let $A_s \times B_t$ be the image of the piece, with A_s equal to $[0,a]$ or $[a,2a]$ according to d_x and B_t equal to $[0,b]$ or $[b,2b]$ according to d_y. Then

$$M^{(p)}_{\alpha\beta}(\boldsymbol{d}) = \frac{1}{ab} \sum_{s,t\in\{-,+\}} \sum_{m=0}^{3}\sum_{n=0}^{1} c_m^s\, e_n^t\, \Delta_{A_s\times B_t} G^{(p)}_{mn}, \tag{55}$$

where, with $G = G^{(p)}_{mn}$, which vanishes on the axes,

$$\begin{aligned}
\Delta_{[0,a]\times[0,b]} G &= G(a,b),\\
\Delta_{[a,2a]\times[0,b]} G &= G(2a,b) - G(a,b),\\
\Delta_{[0,a]\times[b,2b]} G &= G(a,2b) - G(a,b),\\
\Delta_{[a,2a]\times[b,2b]} G &= G(2a,2b) - G(2a,b) - G(a,2b) + G(a,b).
\end{aligned} \tag{56}$$

Every touching moment is therefore an exact, finite combination of $G^{(p)}_{mn}$ at the four points (a,b), $(2a,b)$, $(a,2b)$, $(2a,2b)$. The N-moments follow from Lemma (2.1)(iii) with a and b exchanged.

Proof. By Theorem (4.1), $M^{(p)}_{\alpha\beta}(\boldsymbol{d})$ is the sum over the four pieces $I_s \times I_t$. On a piece substitute $u = \sigma_s a(\nu - d_x)$ and $v = \sigma_t b(\eta - d_y)$: then $\mathrm{d}\nu\,\mathrm{d}\eta = \mathrm{d}u\,\mathrm{d}v/(ab)$, $\rho_{\boldsymbol{d}} = \sqrt{u^2+v^2}$, the weights become the polynomials of Lemma (6.1), and the piece is

Table 2: The weights (23) on the pieces of Lemma (6.1) (x-direction).

d_x	0	0	1	1
piece	I_-, $\sigma=-1$	I_+, $\sigma=+1$	I_-, $\sigma=-1$	I_+, $\sigma=-1$
u	$[0,\,a]$	$[0,\,a]$	$[a,\,2a]$	$[0,\,a]$
W_{00}	$\dfrac{a-u}{a}$	$\dfrac{a-u}{a}$	$\dfrac{2a-u}{a}$	$\dfrac{u}{a}$
W_{10}	$\dfrac{(a-u)^2}{2a^2}$	$\dfrac{(a-u)\,(a+u)}{2a^2}$	$\dfrac{(2a-u)^2}{2a^2}$	$\dfrac{u\,(2a-u)}{2a^2}$
W_{01}	$\dfrac{(a-u)\,(a+u)}{2a^2}$	$\dfrac{(a-u)^2}{2a^2}$	$\dfrac{u\,(2a-u)}{2a^2}$	$\dfrac{u^2}{2a^2}$
W_{11}	$\dfrac{(2a+u)\,(a-u)^2}{6a^3}$	$\dfrac{(2a+u)\,(a-u)^2}{6a^3}$	$\dfrac{(a+u)\,(2a-u)^2}{6a^3}$	$\dfrac{u^2\,(3a-u)}{6a^3}$

$\int_{A_s}\int_{B_t}\sum_{m,n}c_m^s e_n^t u^m v^n (u^2+v^2)^{p/2}\,\mathrm{d}v\,\mathrm{d}u$. Apply (40) term by term; the corners on the axes contribute zero by Proposition (5.1)(i). Each piece is absolutely integrable ($p>-2$), so no limit is involved.

□

Remark. The same corner-sum construction works for every offset, but for distant cells it gives a sum of large, nearly cancelling terms, whereas GL converges exponentially there. It is meant for the touching offsets only.

Corollary 6.1 (Homogeneity). $M^{(p)}_{\alpha\beta}(\boldsymbol{d};\lambda a,\lambda b)=\lambda^p\,M^{(p)}_{\alpha\beta}(\boldsymbol{d};a,b)$ for $\lambda>0$, so the static moments of a grid depend, up to the factor a^p, on the aspect ratio b/a only.

Proof. In (28), $\rho_{\boldsymbol{d}}$ scales by λ and the weights do not depend on a and b.

□

Proposition 6.1 (Agreement with the coincident-cell formulas of the thesis). Consider the integrals $\mathcal{I}^{(-1)}$ and $\mathcal{I}^{(1)}$ of the appendix of [1]. Then

$$\begin{aligned}
(ab)^2 M^{(-1)}_{00}(\mathbf{0}) &= \mathcal{I}^{(-1)}[1],\\
(ab)^2 M^{(-1)}_{11}(\mathbf{0}) &= \mathcal{I}^{(-1)}[\lambda_1\lambda_1'],\\
(ab)^2\big(M^{(-1)}_{10}(\mathbf{0})-M^{(-1)}_{11}(\mathbf{0})\big) &= \mathcal{I}^{(-1)}[\lambda_1(1-\lambda_1')],\\
(ab)^2 M^{(1)}_{00}(\mathbf{0}) &= \mathcal{I}^{(1)}[1],\\
(ab)^2 M^{(1)}_{11}(\mathbf{0}) &= \mathcal{I}^{(1)}[\lambda_1\lambda_1'].
\end{aligned} \tag{57}$$

The first one reads

$$\begin{aligned}
M^{(-1)}_{00}(\mathbf{0}) = \frac{1}{3a^2b^2}\Bigg(&2a^3+2b^3-2\sqrt{a^2+b^2}\,\big(a^2+b^2\big)\\
&+6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)+6ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg).
\end{aligned} \tag{58}$$

Proof. By Corollary (6.1) it suffices to take $a = 1$ and keep $b > 0$ symbolic. The difference of the two sides, written with (55) and the cited formula, simplifies to zero exactly (SymPy).

□

Remark (thesis appendix). The values of the formulas in the appendix of [1] are correct, but three labels are misprinted: the integrands of the $\mathcal{I}^{(1)}$ entries are written with $1/R$ instead of R; the last $\mathcal{I}^{(1)}$ entry repeats the label $[\lambda_1(1-\lambda_1')]$ instead of $[\lambda_2(1-\lambda_2')]$; and the integrand of $\mathcal{I}^{(-1)}[\lambda_2(1-\lambda_2')]$ contains $(u+\lambda_2')$ instead of $(v+\lambda_2')$.

A The mixed antiderivatives $G_{mn}^{(p)}$

This appendix lists the closed forms (50) of the antiderivatives (38) for $p \in \{-1, 1, 3\}$, $m \le 3$ and $n \le 1$, which are the ones Theorem (6.1) needs. Any other one follows from $G_{nm}^{(p)}(b,a) = G_{mn}^{(p)}(a,b)$, and on the axes $G_{mn}^{(p)} = 0$.

Kernel R^{-1}.

$$G_{00}^{(-1)}(a,b) = a\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + b\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)$$

$$G_{01}^{(-1)}(a,b) = \frac{1}{2}\left(-a^2 + a\sqrt{a^2+b^2} + b^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\right)$$

$$G_{10}^{(-1)}(a,b) = \frac{1}{2}\left(-b^2 + b\sqrt{a^2+b^2} + a^2\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\right)$$

$$G_{11}^{(-1)}(a,b) = \frac{1}{3}\left(-a^3 - b^3 + \sqrt{a^2+b^2}\,(a^2+b^2)\right)$$

$$G_{20}^{(-1)}(a,b) = \frac{1}{6}\left(ab\sqrt{a^2+b^2} + 2a^3\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - b^3\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\right)$$

$$G_{21}^{(-1)}(a,b) = \frac{1}{8}\left(-2a^4 + a\sqrt{a^2+b^2}\,(2a^2+b^2) - b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\right)$$

$$G_{30}^{(-1)}(a,b) = \frac{1}{12}\left(2b^4 + b\sqrt{a^2+b^2}\,(a^2-2b^2) + 3a^4\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\right)$$

$$G_{31}^{(-1)}(a,b) = \frac{1}{15}\left(-3a^5 + 2b^5 + \sqrt{a^2+b^2}\,(3a^4 + a^2b^2 - 2b^4)\right)$$

Kernel R^{1}.

$$G_{00}^{(1)}(a,b) = \frac{1}{6}\left(2ab\sqrt{a^2+b^2} + a^3\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\right.$$

$$+ b^3 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{01}^{(1)}(a,b) = \frac{1}{24}\Bigg(-2a^4 + a\sqrt{a^2+b^2}\left(2a^2+5b^2\right) + 3b^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{10}^{(1)}(a,b) = \frac{1}{24}\Bigg(-2b^4 + b\sqrt{a^2+b^2}\left(5a^2+2b^2\right) + 3a^4 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{11}^{(1)}(a,b) = \frac{1}{15}\left(-a^5 - b^5 + \sqrt{a^2+b^2}\left(a^4+2a^2b^2+b^4\right)\right)$$

$$G_{20}^{(1)}(a,b) = \frac{1}{40}\Bigg(ab\sqrt{a^2+b^2}\left(6a^2+b^2\right) + 4a^5 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - b^5 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{21}^{(1)}(a,b) = \frac{1}{144}\Bigg(-8a^6 + a\sqrt{a^2+b^2}\left(8a^4+14a^2b^2+3b^4\right) - 3b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{30}^{(1)}(a,b) = \frac{1}{180}\Bigg(4b^6 + b\sqrt{a^2+b^2}\left(21a^4+2a^2b^2-4b^4\right) + 15a^6 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{31}^{(1)}(a,b) = \frac{1}{105}\left(-5a^7 + 2b^7 + \sqrt{a^2+b^2}\left(5a^6+8a^4b^2+a^2b^4-2b^6\right)\right)$$

Kernel R^3.

$$G_{00}^{(3)}(a,b) = \frac{1}{40}\Bigg(7ab\sqrt{a^2+b^2}\left(a^2+b^2\right) + 3a^5 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 3b^5 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{01}^{(3)}(a,b) = \frac{1}{240}\Bigg(-8a^6 + a\sqrt{a^2+b^2}\left(8a^4+26a^2b^2+33b^4\right)$$

$$+ 15b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{10}^{(3)}(a,b) = \frac{1}{240}\Bigg(-8b^6 + b\sqrt{a^2+b^2}\left(33a^4+26a^2b^2+8b^4\right)$$
$$+ 15a^6 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{11}^{(3)}(a,b) = \frac{1}{35}\Bigg(-a^7-b^7+\sqrt{a^2+b^2}\left(a^6+3a^4b^2+3a^2b^4+b^6\right)\Bigg)$$

$$G_{20}^{(3)}(a,b) = \frac{1}{336}\Bigg(ab\sqrt{a^2+b^2}\left(38a^4+26a^2b^2+3b^4\right)$$
$$+ 18a^7 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 3b^7 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{21}^{(3)}(a,b) = \frac{1}{1920}\Bigg(-48a^8$$
$$+ a\sqrt{a^2+b^2}\left(48a^6+136a^4b^2+118a^2b^4+15b^6\right)$$
$$- 15b^8 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{30}^{(3)}(a,b) = \frac{1}{2240}\Bigg(16b^8$$
$$+ b\sqrt{a^2+b^2}\left(215a^6+134a^4b^2+8a^2b^4-16b^6\right)$$
$$+ 105a^8 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{31}^{(3)}(a,b) = \frac{1}{315}\Bigg(-7a^9+2b^9$$
$$+ \sqrt{a^2+b^2}\left(7a^8+19a^6b^2+15a^4b^4+a^2b^6-2b^8\right)\Bigg)$$

B The touching moments

This appendix lists the x-moments $M_{\alpha\beta}^{(p)}(\boldsymbol{d})$ of the cell $a \times b$ for $p \in \{-1,1,3\}$, $\alpha,\beta \in \{0,1\}$ and the touching offsets $\boldsymbol{d} \in \{(0,0),(1,0),(0,1),(1,1)\}$. They are the corner sums (55) expanded with SymPy, and each one was checked against its corner sum to 30 digits. The N-moments are $N_{\alpha\beta}^{(p)}(d_x,d_y;a,b) = M_{\alpha\beta}^{(p)}(d_y,d_x;b,a)$, and the negative offsets follow from Lemma (2.1)(i) and (ii). For evaluation in double precision

the corner sum (55) is the better form; the expanded formulas are meant for reading and checking.

Kernel R^{-1}

Coincident cell, $\boldsymbol{d} = (0,0)$.

$$M_{00}^{(-1)}(0,0) = \frac{1}{3a^2b^2}\Bigg(2a^3+2b^3-2\sqrt{a^2+b^2}\,(a^2+b^2) + 6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 6ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{10}^{(-1)}(0,0) = \frac{1}{3a^2b^2}\Bigg(a^3+b^3-\sqrt{a^2+b^2}\,(a^2+b^2) + 3a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 3ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{01}^{(-1)}(0,0) = \frac{1}{3a^2b^2}\Bigg(a^3+b^3-\sqrt{a^2+b^2}\,(a^2+b^2) + 3a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 3ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{11}^{(-1)}(0,0) = \frac{1}{90a^4b^2}\Bigg(12a^5+30a^2b^3+2b^5 - \sqrt{a^2+b^2}\,(12a^4+29a^2b^2+2b^4) + 45a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 60a^3b^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared edge parallel to y, $\boldsymbol{d} = (1,0)$.

$$M_{00}^{(-1)}(1,0) = \frac{1}{3a^2b^2}\Bigg(6a^3-b^3+2\sqrt{a^2+b^2}\,(a^2+b^2) - \sqrt{4a^2+b^2}\,(4a^2+b^2) + 12a^2b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) - 6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 6ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) + 6ab^2\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)$$

$$M_{10}^{(-1)}(1,0) = \frac{1}{24a^3b^2}\Bigg(20a^4-8ab^3+2a\sqrt{a^2+b^2}\,(6a^2+11b^2)$$

$$
\begin{aligned}
&- 2a\sqrt{4a^2+b^2}\left(8a^2+7b^2\right)\\
&+ 64a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 40a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 2b^2\left(24a^2-b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ b^2\left(48a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(-1)}(1,0) = \frac{1}{24a^3b^2}\Bigg(&28a^4+2a\sqrt{a^2+b^2}\left(2a^2-3b^2\right)\\
&- 2a\sqrt{4a^2+b^2}\left(8a^2-3b^2\right)\\
&+ 32a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) - 8a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 2b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) + b^4\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(-1)}(1,0) = \frac{1}{360a^4b^2}\Bigg(&180a^5-4b^5+2\sqrt{a^2+b^2}\left(54a^4+13a^2b^2+4b^4\right)\\
&- 2\sqrt{4a^2+b^2}\left(72a^4+11a^2b^2+2b^4\right)\\
&+ 480a^4b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 300a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 30ab^2\left(8a^2+b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 15ab^2\left(16a^2+b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Shared edge parallel to x, $\boldsymbol{d} = (0,1)$.

$$
\begin{aligned}
M_{00}^{(-1)}(0,1) = \frac{1}{3a^2b^2}\Bigg(&-a^3+6b^3+2\sqrt{a^2+b^2}\left(a^2+b^2\right)\\
&- \sqrt{a^2+4b^2}\left(a^2+4b^2\right) - 6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 6a^2 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 12ab^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 6ab^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(-1)}(0,1) = \frac{1}{6a^2b^2}\Bigg(&- a^3 + 6b^3 + 2\sqrt{a^2+b^2}\left(a^2+b^2\right) \\
&- \sqrt{a^2+4b^2}\left(a^2+4b^2\right) - 6a^2 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 6a^2 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 12ab^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 6ab^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(-1)}(0,1) = \frac{1}{6a^2b^2}\Bigg(&- a^3 + 6b^3 + 2\sqrt{a^2+b^2}\left(a^2+b^2\right) \\
&- \sqrt{a^2+4b^2}\left(a^2+4b^2\right) - 6a^2 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 6a^2 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 12ab^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 6ab^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(-1)}(0,1) = \frac{1}{90a^4b^2}\Bigg(&- 6a^5 + 90a^2b^3 + 30b^5 \\
&+ \sqrt{a^2+b^2}\left(12a^4 + 29a^2b^2 + 2b^4\right) \\
&- 2\sqrt{a^2+4b^2}\left(3a^4 + 29a^2b^2 + 8b^4\right) \\
&- 45a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 45a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 120a^3b^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&- 60a^3b^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

Shared vertex, $\boldsymbol{d} = (1,1)$.

$$
\begin{aligned}
M_{00}^{(-1)}(1,1) = \frac{1}{3a^2b^2}\Bigg(& -3a^3 - 3b^3 - 6\sqrt{a^2+b^2}\left(a^2+b^2\right) \\
& + \sqrt{a^2+4b^2}\left(a^2+4b^2\right) + \sqrt{4a^2+b^2}\left(4a^2+b^2\right) \\
& - 12a^2b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 18a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 6a^2b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 12ab^2\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
& + 18ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - 6ab^2\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(-1)}(1,1) = \frac{1}{24a^3b^2}\Bigg(& -10a^4 - 24ab^3 - 2a\sqrt{a^2+b^2}\left(14a^2+39b^2\right) \\
& + 2a\sqrt{a^2+4b^2}\left(3a^2+22b^2\right) \\
& + 2a\sqrt{4a^2+b^2}\left(8a^2+7b^2\right) \\
& - 64a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 104a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 40a^3b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 16b^2\left(6a^2-b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
& + 2b^2\left(72a^2-5b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - b^2\left(48a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(-1)}(1,1) = \frac{1}{24a^3b^2}\Bigg(& -14a^4 - 10a\sqrt{a^2+b^2}\left(2a^2-3b^2\right) \\
& + 2a\sqrt{a^2+4b^2}\left(a^2-6b^2\right) \\
& + 2a\sqrt{4a^2+b^2}\left(8a^2-3b^2\right) \\
& - 32a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 40a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 8a^3b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 16b^4\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) + 10b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - b^4\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(-1)}(1,1) = \frac{1}{360a^4b^2}\Bigg(& -90a^5 - 60b^5 - 6\sqrt{a^2+b^2}\left(42a^4+19a^2b^2+12b^4\right) \\
& + 2\sqrt{a^2+4b^2}\left(27a^4+26a^2b^2+32b^4\right) \\
& + 2\sqrt{4a^2+b^2}\left(72a^4+11a^2b^2+2b^4\right) \\
& - 480a^4b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 780a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 300a^4b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 240ab^2\left(2a^2+b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
& + 30ab^2\left(24a^2+5b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - 15ab^2\left(16a^2+b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Kernel R^1

Coincident cell, $\boldsymbol{d} = (0,0)$.

$$M_{00}^{(1)}(0,0) = \frac{1}{30a^2b^2}\Bigg(2a^5 + 2b^5 - 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) + 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{10}^{(1)}(0,0) = \frac{1}{60a^2b^2}\Bigg(2a^5 + 2b^5 - 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) + 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{01}^{(1)}(0,0) = \frac{1}{60a^2b^2}\Bigg(2a^5 + 2b^5 - 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) + 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{11}^{(1)}(0,0) = \frac{1}{3780a^4b^2}\Bigg(36a^7 + 126a^2b^5 + 8b^7 - \sqrt{a^2+b^2}\left(36a^6 - 165a^4b^2 + 122a^2b^4 + 8b^6\right) + 105a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 210a^3b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared edge parallel to y, $\boldsymbol{d} = (1,0)$.

$$M_{00}^{(1)}(1,0) = \frac{1}{30a^2b^2}\Bigg(30a^5 - b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) - \sqrt{4a^2+b^2}\left(16a^4 - 12a^2b^2 + b^4\right) + 40a^4b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) - 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) + 5ab^4\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)$$

$$M_{10}^{(1)}(1,0) = \frac{1}{720a^3b^2}\Bigg(224a^6 - 24ab^5 + 2a\sqrt{a^2+b^2}\left(16a^4 - 56a^2b^2 + 33b^4\right)$$

$$
\begin{aligned}
&- 2a\sqrt{4a^2+b^2}\left(64a^4-80a^2b^2+21b^4\right)\\
&+ 384a^5b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 84a^5b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 6b^4\left(20a^2-b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 3b^4\left(40a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(1)}(1,0) = \frac{1}{720a^3b^2}\Bigg(&496a^6+2a\sqrt{a^2+b^2}\left(8a^4-16a^2b^2-9b^4\right)\\
&- 2a\sqrt{4a^2+b^2}\left(128a^4-64a^2b^2-9b^4\right)\\
&+ 576a^5b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 36a^5b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 6b^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 3b^6\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(1)}(1,0) = \frac{1}{15120a^4b^2}\Bigg(&3360a^7-16b^7\\
&+ 2\sqrt{a^2+b^2}\left(240a^6-666a^4b^2+55a^2b^4+16b^6\right)\\
&- 2\sqrt{4a^2+b^2}\left(960a^6-960a^4b^2+47a^2b^4+8b^6\right)\\
&+ 5376a^6b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 1176a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 42ab^4\left(20a^2+3b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 21ab^4\left(40a^2+3b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Shared edge parallel to x, $\boldsymbol{d} = (0,1)$.

$$
\begin{aligned}
M_{00}^{(1)}(0,1) = \frac{1}{30a^2b^2}\Bigg(& -a^5 + 30b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) \\
& - \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right) \\
& - 5a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& + 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 5ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(1)}(0,1) = \frac{1}{60a^2b^2}\Bigg(& -a^5 + 30b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) \\
& - \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right) \\
& - 5a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& + 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 5ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(1)}(0,1) = \frac{1}{60a^2b^2}\Bigg(& -a^5 + 30b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) \\
& - \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right) \\
& - 5a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& + 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 5ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(1)}(0,1) = \frac{1}{3780a^4b^2}\Bigg(& -18a^7 + 1890a^2b^5 + 504b^7 \\
& + \sqrt{a^2+b^2}\left(36a^6 - 165a^4b^2 + 122a^2b^4 + 8b^6\right) \\
& - 2\sqrt{a^2+4b^2}\left(9a^6 - 165a^4b^2 + 488a^2b^4 + 128b^6\right) \\
& - 105a^6 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& + 105a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)
\end{aligned}
$$

$$+ 1680a^3b^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$- 210a^3b^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared vertex, $\boldsymbol{d} = (1,1)$.

$$M_{00}^{(1)}(1,1) = \frac{1}{30a^2b^2}\Bigg(- 15a^5 - 15b^5 - 18\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right)$$
$$+ \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right)$$
$$+ \sqrt{4a^2+b^2}\left(16a^4 - 12a^2b^2 + b^4\right)$$
$$- 40a^4b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)$$
$$+ 45a^4b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$
$$- 5a^4b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)$$
$$- 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$+ 45ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)$$
$$- 5ab^4 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)$$

$$M_{10}^{(1)}(1,1) = \frac{1}{720a^3b^2}\Bigg(- 112a^6 - 360ab^5$$
$$- 2a\sqrt{a^2+b^2}\left(80a^4 - 376a^2b^2 + 369b^4\right)$$
$$+ 16a\sqrt{a^2+4b^2}\left(a^4 - 14a^2b^2 + 33b^4\right)$$
$$+ 2a\sqrt{4a^2+b^2}\left(64a^4 - 80a^2b^2 + 21b^4\right)$$
$$- 384a^5b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)$$
$$+ 468a^5b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$

$$
\begin{aligned}
&- 84a^5 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 192b^4\left(5a^2-b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 6b^4\left(180a^2-17b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 3b^4\left(40a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(1)}(1,1) = \frac{1}{720a^3b^2}\Bigg(&- 248a^6 - 34a\sqrt{a^2+b^2}\left(8a^4-16a^2b^2-9b^4\right)\\
&+ 8a\sqrt{a^2+4b^2}\left(a^4-8a^2b^2-18b^4\right)\\
&+ 2a\sqrt{4a^2+b^2}\left(128a^4-64a^2b^2-9b^4\right)\\
&- 576a^5 b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 612a^5 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 36a^5 b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 192b^6\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 102b^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) - 3b^6\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(1)}(1,1) = \frac{1}{15120a^4b^2}\Bigg(&- 1680a^7 - 1008b^7 - 6\sqrt{a^2+b^2}\big(400a^6-1502a^4b^2\\
&+ 269a^2b^4+176b^6\big)\\
&+ 8\sqrt{a^2+4b^2}\left(30a^6-333a^4b^2+110a^2b^4+128b^6\right)\\
&+ 2\sqrt{4a^2+b^2}\left(960a^6-960a^4b^2+47a^2b^4+8b^6\right)\\
&- 5376a^6 b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 6552a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 1176a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&- 1344ab^4\left(5a^2+3b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&+ 126ab^4\left(60a^2+17b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&- 21ab^4\left(40a^2+3b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Kernel R^3

Coincident cell, $\boldsymbol{d} = (0,0)$.

$$
\begin{aligned}
M_{00}^{(3)}(0,0) = \frac{1}{420a^2b^2}\Bigg(&8a^7+8b^7 \\
&- \sqrt{a^2+b^2}\left(8a^6-25a^4b^2-25a^2b^4+8b^6\right) \\
&+ 21a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(3)}(0,0) = \frac{1}{840a^2b^2}\Bigg(&8a^7+8b^7 \\
&- \sqrt{a^2+b^2}\left(8a^6-25a^4b^2-25a^2b^4+8b^6\right) \\
&+ 21a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(3)}(0,0) = \frac{1}{840a^2b^2}\Bigg(&8a^7+8b^7 \\
&- \sqrt{a^2+b^2}\left(8a^6-25a^4b^2-25a^2b^4+8b^6\right) \\
&+ 21a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(3)}(0,0) = \frac{1}{30240a^4b^2}\Bigg(&64a^9+288a^2b^7+16b^9-\sqrt{a^2+b^2}\big(64a^8 \\
&- 275a^6b^2-390a^4b^4+280a^2b^6+16b^8\big) \\
&+ 189a^8 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)
\end{aligned}
$$

$$+ 504a^3b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared edge parallel to y, $\boldsymbol{d} = (1,0)$.

$$\begin{aligned}
M_{00}^{(3)}(1,0) = \frac{1}{420a^2b^2}\Bigg(&504a^7 - 4b^7 \\
&+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right) \\
&- 2\sqrt{4a^2+b^2}\left(128a^6 - 100a^4b^2 - 25a^2b^4 + 2b^6\right) \\
&+ 672a^6b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&- 21a^6b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 21ab^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&+ 21ab^6 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}$$

$$\begin{aligned}
M_{10}^{(3)}(1,0) = \frac{1}{13440a^3b^2}\Bigg(&3936a^8 - 128ab^7 \\
&+ 2a\sqrt{a^2+b^2}\left(80a^6 - 272a^4b^2 - 310a^2b^4 + 177b^6\right) \\
&- 2a\sqrt{4a^2+b^2}\big(1024a^6 - 1152a^4b^2 - 440a^2b^4 \\
&+ 113b^6\big) + 6144a^7b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&- 432a^7b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&- 6b^6\left(112a^2 - 5b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&+ 3b^6\left(224a^2 - 5b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}$$

$$\begin{aligned}
M_{01}^{(3)}(1,0) = \frac{1}{13440a^3b^2}\Bigg(&12192a^8 \\
&+ 2a\sqrt{a^2+b^2}\left(48a^6 - 128a^4b^2 - 90a^2b^4 - 49b^6\right) \\
&- 2a\sqrt{4a^2+b^2}\big(3072a^6 - 2048a^4b^2 - 360a^2b^4
\end{aligned}$$

$$
\begin{aligned}
&\quad - 49b^6\big) + 15360a^7 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&\quad - 240a^7 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 30b^8 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&\quad + 15b^8 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(3)}(1,0) = \frac{1}{120960a^4b^2}\Bigg(&27552a^9 - 32b^9 + 2\sqrt{a^2+b^2}\big(560a^8 - 1702a^6b^2 \\
&- 1590a^4b^4 + 119a^2b^6 + 32b^8\big) \\
&- 2\sqrt{4a^2+b^2}\big(7168a^8 - 7232a^6b^2 - 2280a^4b^4 \\
&+ 103a^2b^6 + 16b^8\big) + 41472a^8 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&- 2916a^8 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&- 18ab^6\left(112a^2+15b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&+ 9ab^6\left(224a^2+15b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Shared edge parallel to x, $\boldsymbol{d} = (0,1)$.

$$
\begin{aligned}
M_{00}^{(3)}(0,1) = \frac{1}{420a^2b^2}\Bigg(&-4a^7 + 504b^7 \\
&+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right) \\
&- 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right) \\
&- 21a^6 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 21a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 672ab^6 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&- 21ab^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$M_{10}^{(3)}(0,1) = \frac{1}{840a^2b^2}\Bigg(-4a^7 + 504b^7$$
$$+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right)$$
$$- 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right)$$
$$- 21a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$
$$+ 21a^6b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)$$
$$+ 672ab^6\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$- 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{01}^{(3)}(0,1) = \frac{1}{840a^2b^2}\Bigg(-4a^7 + 504b^7$$
$$+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right)$$
$$- 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right)$$
$$- 21a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$
$$+ 21a^6b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)$$
$$+ 672ab^6\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$- 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{11}^{(3)}(0,1) = \frac{1}{30240a^4b^2}\Bigg(-32a^9 + 18144a^2b^7 + 4080b^9 + \sqrt{a^2+b^2}\big(64a^8$$
$$- 275a^6b^2 - 390a^4b^4 + 280a^2b^6 + 16b^8\big)$$
$$- 2\sqrt{a^2+4b^2}\big(16a^8 - 275a^6b^2 - 1560a^4b^4$$
$$+ 4480a^2b^6 + 1024b^8\big) - 189a^8b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$

$$
\begin{aligned}
&+ 189a^8 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&+ 16128a^3b^6 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&- 504a^3b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

Shared vertex, $\boldsymbol{d} = (1,1)$.

$$
\begin{aligned}
M_{00}^{(3)}(1,1) = \frac{1}{420a^2b^2}\Bigg(&- 252a^7 - 252b^7\\
&- 33\sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right)\\
&+ 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right)\\
&+ 2\sqrt{4a^2+b^2}\left(128a^6 - 100a^4b^2 - 25a^2b^4 + 2b^6\right)\\
&- 672a^6 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 693a^6 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 21a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 672ab^6 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 693ab^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 21ab^6 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(3)}(1,1) = \frac{1}{13440a^3b^2}\Bigg(&- 1968a^8 - 8064ab^7 - 2a\sqrt{a^2+b^2}\big(1104a^6\\
&- 4880a^4b^2 - 7350a^2b^4 + 7409b^6\big)\\
&+ 16a\sqrt{a^2+4b^2}\left(5a^6 - 68a^4b^2 - 310a^2b^4 + 708b^6\right)\\
&+ 2a\sqrt{4a^2+b^2}\big(1024a^6 - 1152a^4b^2 - 440a^2b^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 113b^6\big) - 6144a^7 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 6576a^7 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 432a^7 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 768b^6\left(28a^2-5b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 6b^6\left(3696a^2-325b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 3b^6\left(224a^2-5b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(3)}(1,1) = \frac{1}{13440a^3b^2}\Bigg(&- 6096a^8\\
&- 130a\sqrt{a^2+b^2}\left(48a^6-128a^4b^2-90a^2b^4-49b^6\right)\\
&+ 16a\sqrt{a^2+4b^2}\left(3a^6-32a^4b^2-90a^2b^4-196b^6\right)\\
&+ 2a\sqrt{4a^2+b^2}\big(3072a^6-2048a^4b^2-360a^2b^4\\
&- 49b^6\big) - 15360a^7 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 15600a^7 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 240a^7 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 3840b^8 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 1950b^8 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 15b^8 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(3)}(1,1) = \frac{1}{120960a^4b^2}\Bigg(&- 13776a^9 - 8160b^9 - 6\sqrt{a^2+b^2}\big(2576a^8\\
&- 10210a^6b^2 - 12690a^4b^4 + 2237a^2b^6 + 1376b^8\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\sqrt{a^2+4b^2}\big(70a^8 - 851a^6b^2 - 3180a^4b^4 \\
&+ 952a^2b^6 + 1024b^8\big) + 2\sqrt{4a^2+b^2}\big(7168a^8 \\
&- 7232a^6b^2 - 2280a^4b^4 + 103a^2b^6 + 16b^8\big) \\
&- 41472a^8 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&+ 44388a^8 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&- 2916a^8 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&- 2304ab^6\left(28a^2+15b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&+ 54ab^6\left(1232a^2+325b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&- 9ab^6\left(224a^2+15b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

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